

drug therapeutic efficacy and reduce side effects.

Environmental applications. Nano solar cells can convert sunlight into electricity more efficiently. These are cheaper to manufacture and easier to install since these can use print-like manufacturing processes and can be made to be flexible.

Nanotechnology-enabled sensors and solutions can now detect and identify chemical or biological agents in the air and soil with much higher sensitivity.

Nanoparticles are being developed to clean industrial water pollutants in groundwater through chemical reactions that render the pollutants harmless.

A nanofabric paper towel woven from tiny wires of potassium-manganese-oxide has been developed. It can absorb 20 times its weight in oil for cleanup applications. Magnetic water-repellent nanoparticles in oil spills use magnets to mechanically remove the oil from the water.

Steps involved in the research of nanotechnology

Jonny Dhiman, research analyst, Promax Legal Solution, says, "Research of nanotechnology is based on the synthesis or analysis of nanoparticles. There are many steps involved in the research of nanotechnology. Two of these are top-down approach and bottom-up approach. In the first, bulk particles are converted into nanoparticles to analyse and study their properties.

"In bottom-up approach, nanoparticles are combined together to analyse the changes in their properties due to structural reformation.

"Steps involved in nanotechnology start with the preparation of materials using techniques so that these are in nano-regime. Then, characterisation of prepared materials is done using various characterisation techniques. Finally, analysis of various physical properties for prepared samples or materials is done."

It is well-known from colloid science that nanoparticles can form aggregates, especially when these are kept as a powder under dry conditions. This aggregation can create difficulties while testing the toxicity of nanoparticles.

Instruments required to study nanoparticles

Atomic force microscopy (AFM) is used for visualising, imaging, measuring and manipulating objects in nanometre scale. Its resolution is of the order of fractions of a nanometre. It has a sharp tip cantilever that is used to scan the surface of the material.

AFM is used to measure many parameters. Some common measurements that are taken are Van der Waals force, chemical bonding, capillary forces and mechanical contact force. Many other parameters can also be measured if additional probes are fitted to the device. This is also known as scanning force microscopy.

A scanning tunnelling microscope is used for imaging surfaces at atomic level. It can be used in different modes like air, water and high vacuum. It can also be used in very low and high temperatures. A difference in voltage is applied when the tip of the device is brought near the material. This causes the electrons to move through the empty space created between them.

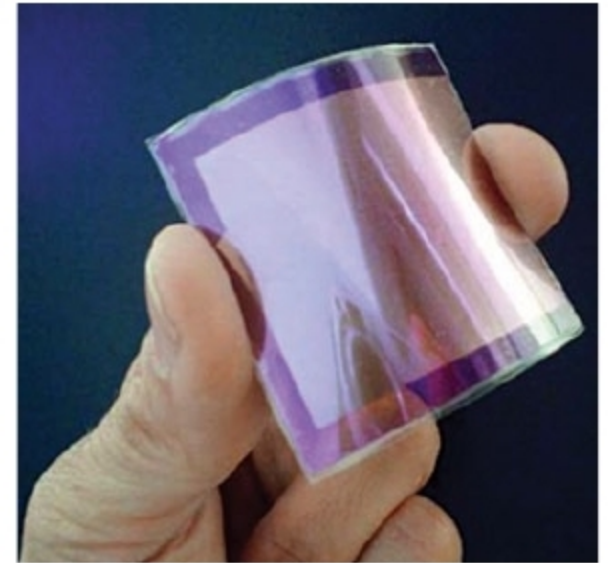
Dhiman adds, "Structural properties of nanoparticles can be studied by x-ray diffraction, morphology by scanning electron microscopy (SEM) or by field emission scanning electron microscope (FESEM), defects by photoluminescence (PL) analysis or by using UV Raman spectroscopy."

Other equipment used in the research of nanoparticles are nanocomposite deposition systems, nano manipulators, nanopositioning systems, nanolithography systems and more.

Available standards

There are various groups involved in developing nanotechnology standards. Some of the leading nanotechnology standards-setting organisations and committees are:

- ISO (or, International Standardization Organization) Technical Committee 229 on Nanotechnologies
- ASTM (or, American Society for Testing and Materials) International Committee E56 Nanotechnology
- IEEE (or, Institute of Electrical and Electronics Engineers) Nanotechnology Council
- International Electrotechnical Commission Technical Commit-



New solar panel films incorporate nanoparticles to create lightweight, flexible solar cells (Credit: Nanosys)

tee 113; it develops nanotechnology standards for electrical and electronic products and systems

Some examples of documentary standards available for nanomaterials are:

- ASTM E2996-15: Guide for workforce education in nanotechnology health and safety
- ASTM E2490-09(2015): Standard test method for analysis of hemolytic properties of nanoparticles
- ASTM E2524-08(2013): Standard test method for analysis of hemolytic properties of nanoparticles
- ASTM E2909-13: Standard file format for the submission and exchange of data on nanomaterials
- ISO/TR 13014:2012: Physico-chemical characterisation of engineered nanoscale materials for toxicologic assessment

To sum up

Nanotechnology is an emerging field of science and technology. It is a field of research and innovation concerned with building materials and devices on atomic scale. Ordinary rules of physics and chemistry at such scales no longer apply. Basically, here physics leaves its classical realm and enters a new world of quantum mechanics.

The scientific community, policymakers, apex bodies and funding agencies are continuously striving to progress in various areas of nanoscience and nanotechnology. In the last two decades, India has achieved many milestones in the areas of nanoelectronics, nanomaterials and nano-biotechnology, and many are yet to be achieved. **EFY**